Guarantees for kernel herding: ball approximation and convergence rate on the unit cube

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned source file sources/ahl-271-compressed-empirical-measures-in-finite-dimensions-2026/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. References inside the span are handed to the assembler as navigation stubs (same-article results the counted proof cites); each stub records non_load_bearing and the reason. Stacks tag stubs are not generated per the S062 chapter config (claim_prefix doi:10.5802/ahl, tag index never consulted).

Editorial note. Printed number of the counted statement in the original: 4.1 (cedram shared theorem counter, section 4); the card reproduces it with initial_counters = {"section": 4, "theo": 0}. The unit's census v3.17 number was drifted for this item; this number was verified against the publisher PDF (ahl.centre-mersenne.org/item/10.5802/ahl.271.pdf) by matching the statement text verbatim.

Editorial note. Citations (none) are neutralized by the no-bibliography card convention.

Editorial note. Judgment note (this lane's acceptance): complete author derivation inside the span; see lane-s11/ahl-271_faces.txt and the deep-read ledger (ahl-271 prop:KH_guarantees, printed 4.1, PDF-verified).

Editorial scope. Single proposition card cut from the AHL article doi:10.5802/ahl.271 (ahl-271); statement and proof are the author's own text at source lines 2903-2930 and 2931-2976.

Allowed standard prerequisites (editorial).

- Lipschitz condition on the RKHS (equation def:Lip_for_RKHS, printed (2.13)). cited inside the counted statement/proof spans. the Lipschitz hypothesis used throughout
- Empirical ball theorem (Theorem thm:ball_in_empirical, printed 2.16). cited inside the counted statement/proof spans. the source of the ball statement

Source reference. Original definition 2.13: sources/ahl-271-compressed-empirical-measures-in-finite-dimensions-2026/pinned-original-source.tex, source lines 1481–1481. navigation stub only: the target lives outside the counted span in the same article; the counted proof reads the cited result/identity from the stub text, so the stub is not load-bearing

Source reference. Original theorem 2.16: sources/ahl-271-compressed-empirical-measures-in-finite-dimensions-2026/pinned-original-source.tex, source lines 1704–1704. navigation stub only: the target lives outside the counted span in the same article; the counted proof reads the cited result/identity from the stub text, so the stub is not load-bearing

Proposition 4.1. *Let $X_1, \ldots, X_n$ be i.i.d. random variables on some probability space $(\Omega, \mathcal{A}, P)$, which attain values in $\mathcal{X} = [0,1]^l, l \geq 1$, and let k be a continuous kernel function on $\mathcal{X}$ such that the corresponding RKHS $\mathcal{H}$ is d-dimensional, $1 \leq d < \infty$, functions $h \in \mathcal{H}$ are Lipschitz continuous in the sense of* (2.13) *with Lipschitz constant $L > 0$, and $\mathbf{1} \in \mathcal{H}$. Furthermore, assume that the law of X_1 has a density p on $\mathcal{X}$ and $\inf_{x \in \mathcal{X}} p\ (x) \geq c > 0$ for some constant c. Let $\widetilde{\lambda}_d$ be the smallest eigenvalue of the Mercer decomposition. With probability $1 - q, q \in (0,1)$,*

$$\left\| \widehat{\mathfrak{m}}_{\lceil n^{1/2} \rceil} - \mathfrak{m}_n \right\| \leq \frac{32 \|k\|_\infty}{\delta}\, n^{-1/2}$$

whenever

$$n > \left(\frac{\sqrt{2 \log(4/q)} + 96 \|k\|_\infty^{1/2} / \delta}{c \beta_l (\delta / 8L)^l} \right)^2 \vee \left(\frac{\sqrt{2} \|k\|_\infty^{1/2} (1 + \sqrt{\log(2/q)})}{\delta / 4} \right)^2$$

and where

$$\delta = \frac{c(\widetilde{\lambda}_d^{1/2} \wedge L)^{l+1} \beta_l}{(l+1)(2L)^l}.$$

Furthermore, for any $t \geq 1$ and $n \geq 1$,

$$E(\|\widehat{\mathfrak{m}}_t - \mathfrak{m}_n\|) \leq 32 \|k\|_\infty / t\delta + 4 \|k\|_\infty^{1/2} \exp\bigl(-(c_1 n^{1/2} - c_2)_+^2\bigr) / t^{1/2},$$

where c_1 and c_2 can be chosen as

$$c_1 = \frac{\delta}{\sqrt{32}\|k\|_\infty^{1/2}} \wedge c\beta_l(\delta/8L)^l \quad \textit{and} \quad c_2 = 1 \vee 96\|k\|_\infty^{1/2}/\delta$$

If the stopping criterion of the algorithm is an error of $\|\widehat{\mathfrak{m}}_{\hat{t}} - \mathfrak{m}_n\| \leq n^{-1/2}$, i.e. $\widehat{t} = \inf_{t \geq 1} \|\widehat{\mathfrak{m}}_t - \mathfrak{m}_n\| \leq n^{-1/2}$, and if the infimum is greater than n, then $\widehat{t} = n$ and $\widehat{\mathfrak{m}}_{\hat{t}} = \mathfrak{m}_n$, then

$$E(\widehat{t}) \leq \left\lceil 32\|k\|_\infty n^{1/2}/\delta \right\rceil + 2n\exp\bigl(-(c_1 n^{1/2} - c_2)_+^2\bigr)$$

whenever $n \geq N$.

Proof. The *first statement* follows directly from Theorem 2.16 and the bound on the error of the kernel herding algorithm. For the *second statement*, observe that

$$\Pr(\|\mathfrak{m}_n - \mathfrak{m}\| \geq \delta/4) \leq \exp\left(-\left(\frac{\delta n^{1/2}}{\sqrt{32}\|k\|_\infty^{1/2}} - 1\right)^2\right)$$

whenever $n \geq (\sqrt{32}\|k\|_\infty^{1/2}/\delta)^2$, follows by the same argument as in Theorem 2.16. Similarly, there is a ball of radius $\delta/2$ around $\mathfrak{m}$ in C_n in the affine span of C_n with probability at least

$$1 - (1/2)\exp\Bigl(-\bigl(n^{1/2}c\beta_l(\delta/8L)^l - 96\|k\|_\infty^{1/2}/\delta\bigr)^2\Bigr)$$

whenever $n \geq (96(8L)^l\|k\|_\infty^{1/2}/c\beta_l\delta^{l+1})^2$. In particular, at the event that there also exists a ball of radius $\delta/2$ around $\mathfrak{m}$ and $\|\mathfrak{m}_n - \mathfrak{m}\| < \delta/4$ we have a ball of radius $\delta/4$ around $\mathfrak{m}_n$ in C_n and $\|\widehat{\mathfrak{m}}_t - \mathfrak{m}_n\| \leq 32\|k\|_\infty/\delta t$. The probability that either $\mathfrak{m}_n$ is not $\delta/4$ close to $\mathfrak{m}$ or that there is no ball of radius $\delta/2$ around $\mathfrak{m}$ in C_n is upper bounded by

$$\exp\left(-\left(\frac{\delta n^{1/2}}{\sqrt{32}\|k\|_\infty^{1/2}} - 1\right)^2\right) + (1/2)\exp\Bigl(-\bigl(n^{1/2}c\beta_l(\delta/8L)^l - 96\|k\|_\infty^{1/2}/\delta\bigr)^2\Bigr). \tag{1}$$

To simplify the bound consider the expressions for the constants c_1, c_2 in the theorem statement

$$c_1 = \frac{\delta}{\sqrt{32}\|k\|_\infty^{1/2}} \wedge c\beta_l(\delta/8L)^l \quad \text{and} \quad c_2 = 1 \vee 96\|k\|_\infty^{1/2}/\delta.$$

The constant c_1 is simply the minimum of the left hand terms in the two exponentials in Eq. (1) and c_2 the maximum of the right hand terms. Using these we can bound the sum of the two exponentials as long as $c_1 \geq c_2$. We deal with the case that $c_2 > c_1$ by using the operator $(a)_+ = a \vee 0$ in the following expression upper bound on Eq. (1)

$$2\exp\Big(-(c_1 n^{1/2} - c_2)_+^2\Big)$$

Notice that even when there is no ball around $\mathfrak{m}_n$, for $t \geq 1$ it holds that

$$\|w_{t+1}\|^2 = \|k(X_{\iota(t)}, \cdot\,) - \mathfrak{m}_n\|^2 + \|w_t\|^2 - 2\big\langle k(X_{\iota(t)}, \cdot\,) - \mathfrak{m}_n, \mathfrak{m}_n\big\rangle.$$

Now, $\langle k(X_{\iota(t)}, \cdot\,), \mathfrak{m}_n\rangle \geq \|\mathfrak{m}_n\|^2$ since $\mathfrak{m}_n$ lies in C_n which has extremes $k(X_1, \cdot\,), \ldots,$
$k(X_n, \cdot\,)$, and

$$t^2\|\widehat{\mathfrak{m}}_t - \mathfrak{m}_n\|^2 = \|w_{t+1}\|^2 \leq \sum_{i=1}^{t}\big\|k(X_{\iota(t)}, \cdot\,) - \mathfrak{m}_n\big\|^2 \leq 4t\|k\|_\infty.$$

Hence, in the event that there is no ball we still have that $\|\widehat{\mathfrak{m}}_t - \mathfrak{m}_n\| \leq 2t^{-1/2}\|k\|_\infty^{1/2}$.

Combining the bound for the case that there is a ball with the bound for the case that there is no ball, we find that

$$E(\|\widehat{\mathfrak{m}}_t - \mathfrak{m}_n\|) \leq 32\|k\|_\infty/t\delta + 4\|k\|_\infty^{1/2}\exp\big(-(c_1 n^{1/2} - c_2)_+^2\big)t^{-1/2}$$

whenever n is large enough and with c_1, c_2 as in the theorem statement.

The *third statement* follows along similar lines. In the event that we have a ball of size $\delta/4$ it follows that $\|\widehat{\mathfrak{m}}_t - \mathfrak{m}_n\| \leq 32\|k\|_\infty/t\delta$. Setting the right hand side to $n^{-1/2}$ leads to $\widehat{t} \leq \lceil 32\|k\|_\infty n^{1/2}/\delta\rceil$. If this event does not occur then $\|\widehat{\mathfrak{m}}_t - \mathfrak{m}_n\| \leq 2\|k\|_\infty^{1/2}t^{-1/2}$ and $\widehat{t} \leq \lceil 4\|k\|_\infty n\rceil$, but the algorithm stops when $\widehat{t} > n$ and the trivial upper bound $\widehat{t} \leq n$ is more useful. Combining these we find that

$$E(\widehat{t}) \leq \Big\lceil 32\|k\|_\infty n^{1/2}/\delta\Big\rceil + 2n\exp\big(-(c_1 n^{1/2} - c_2)_+^2\big)$$

when n is large enough. □